ثانوية المدينة التعليمية
Education City High School

2.9

Logarithmic Expressions

Student Notes and Guided Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Translate between exponential and logarithmic form.
- Evaluate logarithmic expressions exactly when possible.
- Approximate logs with technology and interpret them in scale contexts.

Essential Question

How does a logarithm answer an exponent question?

$$b^y = x \xleftrightarrow{\text{same fact}} \log_b x = y$$

$b > 0,\ b \neq 1$ $\quad\quad$ $x > 0$

Concept Framework

Key Idea

$\log_b x$ is the exponent to which b must be raised to obtain x.

Key Idea

Logarithm inputs must be positive.

Key Idea

Base must be positive and not 1.

Key Idea

Exact evaluation is often easiest by rewriting as an exponential equation.

Formula Map

Formula and Structure

$$\log_b x = y \iff b^y = x$$

Formula and Structure

$$\log_b 1 = 0$$

Formula and Structure

$$\log_b b = 1$$

Worked Examples

Worked Example 1

Problem. Evaluate $\log_3(1/27)$.
Solution. Since $3^{-3} = 1/27$, the value is -3.

Worked Example 2

Problem. Solve $\log_5 x = 3/2$.
Solution. $x = 5^{3/2} = 5\sqrt{5}$.

Worked Example 3

Problem. Explain why $\log_2(-8)$ is not real.
Solution. No real power of positive base 2 produces a negative output.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Evaluate exactly: $\log_2 32$.

2. Evaluate exactly: $\log_3(1/27)$.

3. Evaluate exactly: $\log_{10} 0.001$.

4. Evaluate exactly: $\ln e^7$.

5. Evaluate exactly: $\log_4 8$.

6. Solve $\log_5 x = -2$.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is ____________________.

2. A restriction I must remember is ____________________.

3. One graphical feature I can justify algebraically is ____________________.

Video support: Flipped Math topic page | Original ECHS material informed by Pearson and Holt Precalculus concepts.